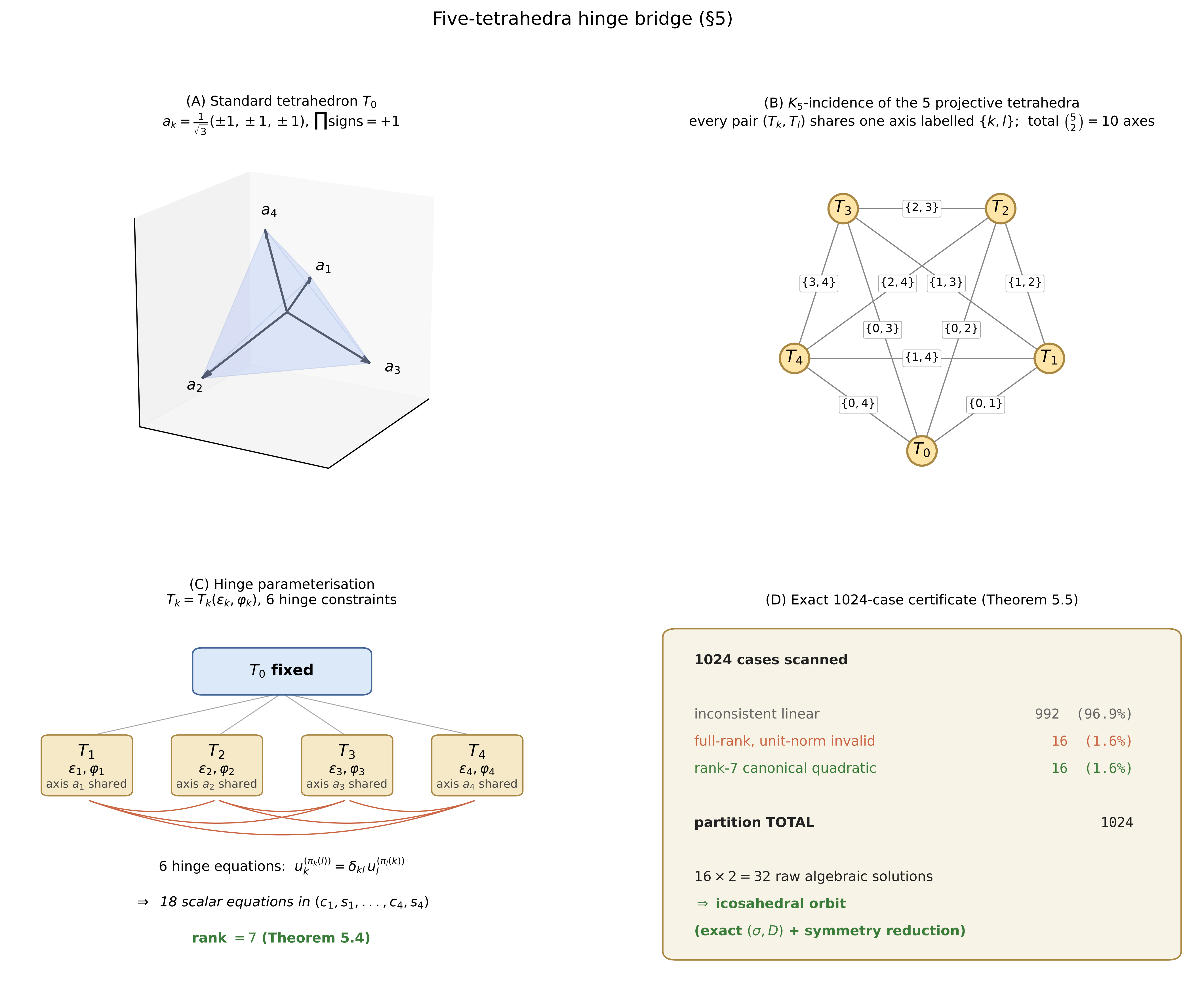

Five-tetrahedra hinge bridge (§5)
(A) Standard tetrahedron T_0
$a_k = \frac{1}{\sqrt{3}}(\pm 1, \pm 1, \pm 1)$, $\prod$ signs $= +1$
a_4
a_1
a_3
a_2
(B) K_5-incidence of the 5 projective tetrahedra
every pair (T_k, T_l) shares one axis labelled $\{k, l\}$; total $\binom{5}{2} = 10$ axes
T_3
$\{2,3\}$
T_2
$\{3,4\}$
$\{2,4\}$
$\{1,3\}$
$\{1,2\}$
$\{0,3\}$
$\{0,2\}$
T_4
$\{1,4\}$
T_1
$\{0,4\}$
$\{0,1\}$
T_0
(C) Hinge parameterisation
$T_k = T_k(\varepsilon_k, \varphi_k)$, 6 hinge constraints
T_0 **fixed**
T_1 ε_1, φ_1 axis a_1 shared
T_2 ε_2, φ_2 axis a_2 shared
T_3 ε_3, φ_3 axis a_3 shared
T_4 ε_4, φ_4 axis a_4 shared
6 hinge equations: $u_k^{(\pi_k(l))} = \delta_{kl}\, u_l^{(\pi_l(k))}$
$\Rightarrow$ *18 scalar equations in* $(c_1, s_1, ..., c_4, s_4)$
rank $= 7$ **(Theorem 5.4)**
(D) Exact 1024-case certificate (Theorem 5.5)
1024 cases scanned
inconsistent linear 992 (96.9%)
full-rank, unit-norm invalid 16 (1.6%)
rank-7 canonical quadratic 16 (1.6%)
partition TOTAL 1024
$16 \times 2 = 32$ raw algebraic solutions
$\Rightarrow$ **icosahedral orbit**
(exact (σ, D) **+ symmetry reduction)**